

## QUANTUM OPTICS

## Topological Hong-Ou-Mandel interference

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The interplay of topology and optics provides a route to pursue robust photonic devices, with the application to photonic quantum computation in its infancy. However, the possibilities of harnessing topological structures to process quantum information with linear optics, through the quantum interference of photons, remain largely uncharted. Here, we present a Hong-Ou-Mandel interference effect of topological origin. We show that this interference of photon pairs—ranging from constructive to destructive—is solely determined by a synthetic magnetic flux, rendering it resilient to errors on a fundamental level. Our implementation establishes a quantized flux that facilitates exclusively destructive quantum interference. Our findings pave the way toward the development of next-generation photonic quantum circuitry and scalable quantum computing protected by virtue of topologically robust quantum gates.

Topological quantum photonics combines two emerging fields of physics: In one of these, topological states in condensed-matter physics with their robust wave transport have inspired efforts to apply this universal concept for wave systems to electromagnetics (1), photonics (2, 3), acoustics (4), cold atoms (5, 6) and mechanics (7). By contrast, quantum photonics uses interference (8) and entanglement (9, 10) among many-photon states for quantum computation and quantum information processing (11, 12). As the scalability of the latter is limited by scattering losses and the detrimental impact of manufacturing imperfections, bringing topological features to bear on photonic quantum circuits is a particularly promising approach to establish robustness in photonic quantum computing (13). The first demonstrations of topological transport on edge channels of single photons (14) were soon followed by proposals to use similar principles to safeguard delicate biphoton properties such as correlations (15) and entanglement (16). Furthermore, it has been experimentally shown that even one-dimensional topological systems outperform their conventional counterparts (17, 18). It is also recognized that chiral quantum-optic interfaces (19) may serve as topological sources of distinguishable (20) and indistinguishable photon pairs (21) with inherent robustness in spectral correlations between the photon pairs. Beyond the synthesis and transport of such states, Hong-Ou-Mandel (HOM) interference, two-photon interference in a beam splitter or directional coupler leading to the suppression of coincidences among the output of the photonic device (8, 22, 23), is a fundamental building block for linear optical quantum computing.

Here, we introduce topological HOM interference equipped with intrinsic topological

protection against imperfections. Specifically, we apply Aharonov-Bohm (AB) interference of a single particle (24) to photon pairs, such that their interference is solely dependent on the magnetic flux and the associated Berry phase (25, 26). The hallmark two-photon suppression occurs at a flux of  $\pi$  and has to be implemented by means of artificial gauge fields, because photons as uncharged particles are immune to external fields. We tailor the birefringence of waveguides in a directional coupler to leverage the additional dynamics in the polarization degree of freedom to implement artificial gauge fields with a quantized magnetic flux of  $\pi$ . Note that in momentum space (27), this relates to a Chern number (28) as topological invariant. Consequently, coincidences of indistinguishable photons are fully suppressed regardless of the length of the directional coupler. This is in stark contrast to conventional designs wherein the visibility of quantum interference periodically oscillates upon propagation (29, 30). Further we demonstrate experimentally that our topological architecture protects the HOM interference against perturbations such as impurities of the input states or deviations in coupling strengths.

### Photonic system—theory and experimental platform

Our system is a square plaquette (in the following symbolized as  $\square$ , with Hamiltonian  $\hat{\mathcal{H}}_{\square}$ ) of four sites with equal couplings  $C_{j,k}$  between the pairs of sites  $(j, k) \in \mathcal{M} = \{(1, 2), (1, 3), (2, 4), (3, 4)\}$  that is permeated by a flux  $\Phi$  by virtue of an artificial magnetic field (Fig. 1A). The propagation of a single photon along the  $z$  direction, created on site  $j$  with respective creation operator  $\hat{a}_j^\dagger$ , is then described by the Heisenberg equation of motion

$$i \frac{\partial}{\partial z} \hat{a}_j^\dagger = [\hat{a}_j^\dagger, \hat{\mathcal{H}}_{\square}],$$

$$\hat{\mathcal{H}}_{\square} = C \sum_{(j,k) \in \mathcal{M}} e^{i\varphi_{j,k}} \hat{a}_j^\dagger \hat{a}_k + \text{h.c.} \quad (1)$$

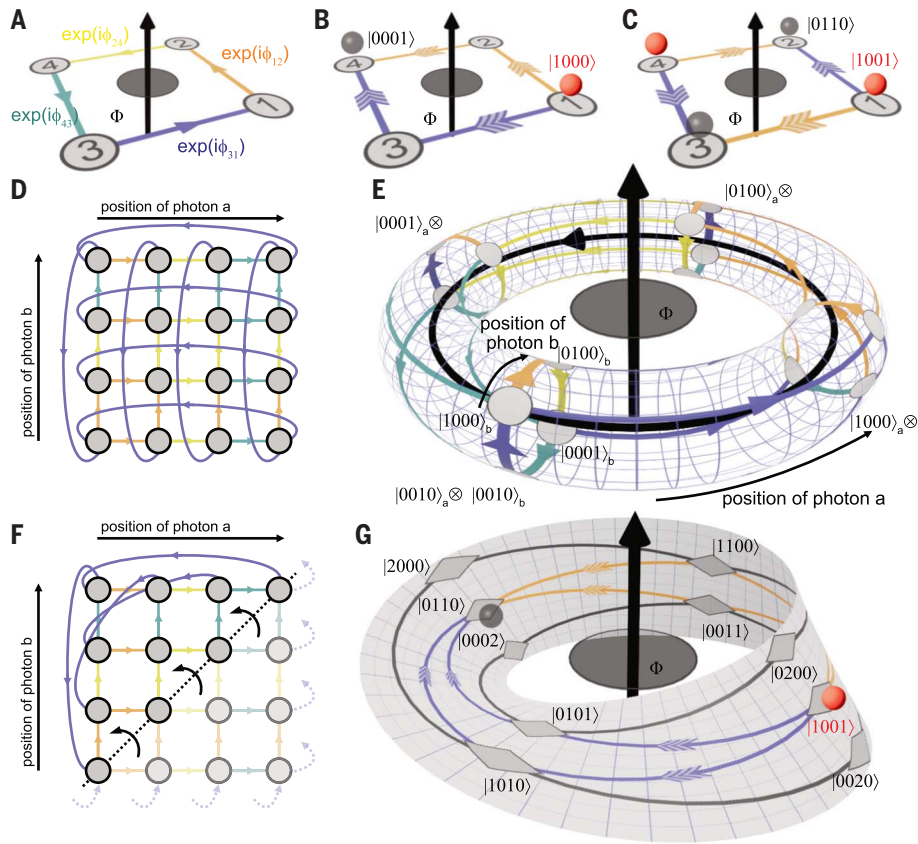
where h.c. denotes the Hermitian conjugate and  $\varphi_{j,k} = \arg C_{j,k}$  are the arguments of the complex-

valued coupling strengths  $|C_{j,k}| = C$ . In the Peierls' substitution (31) gives rise to the  $\Phi = \oint_{\square} d\varphi$  enclosed by the plaquette. The resulting single-photon propagation is closely related to the evolution of an electron in the AB effect (Fig. 1B): Single photons with the initial state  $\hat{a}_1^\dagger |\mathbf{0}\rangle = |1000\rangle$  propagate clockwise (blue path) and counterclockwise (orange path) around the flux-threaded plaquette and interfere at site  $j = 4$  (gray sphere) with flux-induced phase differences that are accumulated along the different paths, where  $|\mathbf{0}\rangle$  denotes the multimode vacuum state. Thus, the probability of finding the photon on site  $j = 4$  scales with  $\propto \cos^2 \frac{\Phi}{2}$ , in particular establishing constructive (destructive) interference at  $\Phi = 0$  ( $\Phi = \pi$ ), respectively, as has been recently demonstrated with classical laser light in an integrated photonic setting (32, 33). We apply this effect to a two-photon evolution on the plaquette. Figure 1C illustrates the contributing single-photon trajectories (blue and orange arrows) that convert the initial two-photon state  $\hat{a}_1^\dagger \hat{a}_4^\dagger |\mathbf{0}\rangle = |1001\rangle$  into  $\hat{a}_2^\dagger \hat{a}_3^\dagger |\mathbf{0}\rangle = |0110\rangle$ . The underlying principle behind the considered two-photon interference is contained in the transition from distinguishable to indistinguishable photons: The twofold Hamiltonian of two distinguishable photons  $a$  and  $b$ ,  $\hat{\mathcal{H}}_2 = \hat{\mathcal{H}}_{\square}^a \oplus \hat{\mathcal{H}}_{\square}^b$ , on the square plaquette can be illustrated by the mesh shown in Fig. 1D. Notably, this resembles a torus (Fig. 1E) where the toroidal and poloidal coordinates encode the positions of photons  $a$  and  $b$  on the square plaquette. Crucially, each photon is exclusively affected by the specific flux that it is subjected to. In the transition from distinguishable to indistinguishable photons depicted in Fig. 1F, two-photon states, which are identical under the photon's permutation, are removed by folding the mesh along the line of states in which the photons coincide. The resulting two-photon Hamiltonian (Fig. 1G) instantiates a Möbius strip (34) that encloses the same flux  $\Phi$  (35) as the square plaquette does for single photons [details provided in the supplementary text (36)]. Furthermore, the single-photon AB effect depicted in Fig. 1B is repeated in the two-photon trajectories that transfer to clockwise (blue) and counterclockwise (orange) trajectories around the Möbius strip (Fig. 1G). Hence, the interference of the two paths dictates a scaling of the probability of finding the photons in the state  $|0110\rangle$  with  $\propto \cos^2 \frac{\Phi}{2}$ , and, in particular, allows for this state to be suppressed by means of destructive quantum interference when  $\Phi = \pi$ , whereas distinguishable photons can still be found in this configuration.

Next, we turn to the implementation of a quantized  $\Phi = \pi$  flux on a square plaquette as basis for error-resilient HOM interference. To this end, we use a birefringent directional coupler (BDC) composed of two waveguides with, in principle, independently customizable principal axis orientations as well as strengths of



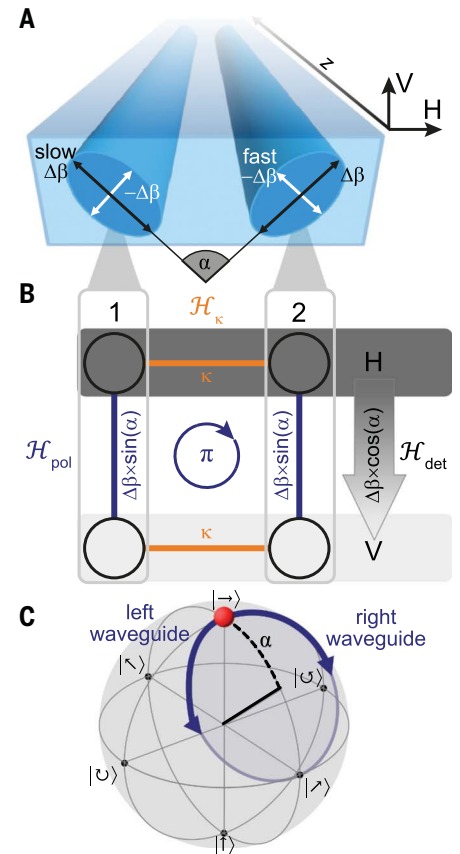
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**Fig. 1. The two-particle Aharonov-Bohm effect on a square plaquette.** (A) Schematic illustration of a square plaquette as a Hamiltonian with complex hopping rates. According to Peierls' substitution, the integral of the coupling rate's arguments on the closed loop gives rise to the magnetic flux  $\Phi$  shown in the center. (B) A single particle in the initial state  $\hat{a}_1^\dagger|0\rangle = |1000\rangle$  (red sphere) at site 1 of a square plaquette encloses a magnetic flux  $\Phi$  with its clockwise and counterclockwise trajectories (marked blue and orange via site 3 and 2, respectively) on the way to its final state  $\hat{a}_4^\dagger|0\rangle = |0001\rangle$  (gray sphere). (C) Two photons launched in the initial state  $\hat{a}_1^\dagger\hat{a}_4^\dagger|0\rangle = |1001\rangle$  at the sites 1 and 4 (red spheres) can take either the blue or orange marked trajectories to the final state  $\hat{a}_2^\dagger\hat{a}_3^\dagger|0\rangle = |0110\rangle$ , where the photons are found on sites 2 and 3, respectively. (D) The Hamiltonian for distinguishable biphotons as a twofold single-photon Hamiltonian  $\hat{\mathcal{H}}_2 = \hat{\mathcal{H}}_\oplus^a \oplus \hat{\mathcal{H}}_\oplus^b$  resembles (E), a torus with the position of a photon a and b as a toroidal and poloidal coordinate, respectively. The color scheme of the connections corresponds to that in (A). (F) The mesh of the torus is converted to the Hamiltonian for indistinguishable photons. Hence, sites, that are identical under photon exchange are removed by folding the twofold Hamiltonian along the line of its orbifold points (points of coinciding single photons). (G) Reconnecting the mesh in the same manner as the torus, the connectivity pattern instantiates a Möbius strip threaded with a magnetic flux  $\Phi$  through the center, where each site (diamond) and each connection represents one two-particle configuration and connection of an individual particle on the square plaquette, respectively. The marked paths in (C) now represent clockwise (blue) and counterclockwise (orange) paths from initial to final two-particle states on a connectivity tree shaped as a Möbius strip threaded with a magnetic flux  $\Phi$  through the center.

birefringence. For the sake of simplicity, we consider waveguides with identical birefringence  $\Delta\beta = (\beta_t - \beta_s)/2$  (Fig. 2A). To establish single-photon dynamics in the polarization degree of freedom (37), a horizontal and vertical (H and V) reference frame is chosen such that the slow axes are oriented at angles of  $\pm\alpha/2$  with respect to the V axis. Consequently, the single-photon Hamiltonian  $\hat{\mathcal{H}}_{\text{BDC}} = \hat{\mathcal{H}}_\kappa + \hat{\mathcal{H}}_{\text{pol}} + \hat{\mathcal{H}}_{\text{det}}$  comprises evanescent coupling  $\hat{\mathcal{H}}_\kappa$  with a coupling rate  $\kappa$  (similar to a conventional directional coupler) and polarization cou-

pling  $\hat{\mathcal{H}}_{\text{pol}} \propto \pm \Delta\beta \sin \alpha$  (sign changes among the waveguides) as well as polarization detuning  $\hat{\mathcal{H}}_{\text{det}} \propto \Delta\beta \cos \alpha$  in each waveguide [see supplementary text for details (36)]. The joint dynamics (Fig. 2B) mediated by  $\hat{\mathcal{H}}_\kappa + \hat{\mathcal{H}}_{\text{pol}}$  give rise to a connectivity pattern in a rectangularly shaped plaquette, whereas the polarization detuning  $\hat{\mathcal{H}}_{\text{det}}$  can be interpreted as a scalar electric potential of strength  $\Delta\beta \cos \alpha$ , i.e., an artificial electric field that vanishes for perpendicular slow axes. Crucially,  $\hat{\mathcal{H}}_{\text{pol}}$  entails real-valued coupling rates with opposite sign. Hence, the  $\pi$  difference



**Fig. 2. Single-photon dynamics in a birefringent directional coupler.** (A) Two identical, birefringent waveguides with slow axes rotated clockwise and counterclockwise by  $\pm\alpha/2$  with respect to the vertical reference. Therefore, the slow axes intersect at an angle  $\alpha$ . (B) The resulting single-photon dynamics in this waveguide system are driven by the evanescent coupling ( $\hat{\mathcal{H}}_\kappa$ , orange) with the rate  $\kappa$  between the two waveguides (1, 2), the polarization coupling ( $\hat{\mathcal{H}}_{\text{pol}}$ , blue) with rate  $\Delta\beta \sin \alpha$ , and the detuning ( $\hat{\mathcal{H}}_{\text{det}}$ ) between horizontal (H) and vertical (V) polarizations with the rate  $\Delta\beta \cos \alpha$ . (C) Depending on which waveguide it occupies, the polarization of an initially horizontally polarized photon launched in the BDC performs a left- or right-handed rotation on the Poincaré sphere around an axis that is located in the  $S_1$ - $S_2$  plane. The rotation axis and the  $S_1$  axis enclose the angle  $\alpha$ .

in the arguments in the polarization coupling establishes the  $\Phi = \pi$ -flux when combined with the evanescent coupling rates with  $\arg \kappa = 0$  of  $\hat{\mathcal{H}}_\kappa$ . The manifestation of the  $\Phi = \pi$ -flux in the BDC is illustrated in Fig. 2C: During its propagation, a horizontally polarized photon rotates around one axis in the  $S_1$ - $S_2$  plane (orientation given by  $\alpha$ ) on the Poincaré sphere. Whether the rotation is left- or right-handed depends on which waveguide the photon occupies. The opposite handedness does not change with  $\alpha$ , unless the trajectories shrink into a single point for

$\alpha = 0^\circ$  ( $180^\circ$ ). Thus, the artificial magnetic flux over a full roundtrip is quantized as

$$\Phi = \begin{cases} 0 & \alpha = 0^\circ, \\ \pi & \text{otherwise} \end{cases}$$

Note that in the case  $\alpha = 0^\circ$  (both principal axes parallel to the input state), in which no polarization oscillation occurs in the H/V reference frame, arbitrarily rotated reference frames provide rectangularly shaped plaquettes without flux (37) as an exception from the otherwise always present flux of  $\Phi = \pi$ .

### Propagation-invariant coincidence suppression

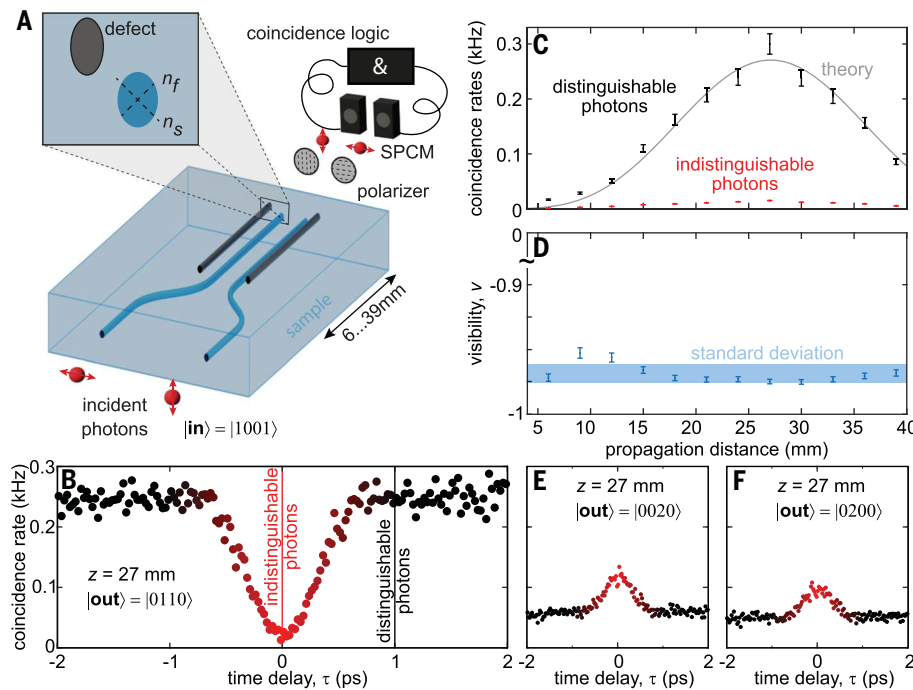
We fabricated BDCs (Fig. 3A) by means of femtosecond laser direct writing in fused silica (38, 39). Each waveguide exhibits birefringence (40) with principal axes aligned to the horizontal and vertical edges of the sample. Thus, the horizontally and vertically polar-

ized incident photons maintain their polarization as the waveguides converge from an initial separation of 127  $\mu\text{m}$  (effectively zero coupling) to the eventual separation of 29.5  $\mu\text{m}$  within the interaction region that stretches until the end of the sample [see materials and methods for details on the implementation (36)]. Simultaneously, the principal axes of the left and right waveguide are rotated in opposite directions by about  $\pm(45.3 \pm 0.7)^\circ$  by inscribing heavily detuned defect lines with slower inscription speed in close proximity to the original waveguides. Owing to the strong detuning, coupling of photons to these defects is negligible, allowing their stress fields to be utilized to modify the birefringent properties of the waveguides as a function of their relative position (41). The perpendicular principal axis alignment among the two waveguides ( $\alpha = 90^\circ$ ) yields a vanishing detuning  $\mathcal{H}_{\text{det}} \propto \cos \alpha$  and subjects the photons solely to the artificial magnetic flux  $\Phi = \pi$ . Moreover, is-

sues of polarization-dependent coupling rates (42) vanish because coupling exclusively occurs among fast and slow axes of different waveguides [for details, see the supplementary text (36)]. To realize a square plaquette, we match the evanescent coupling strength  $\kappa = (0.043 \pm 0.002) \text{ mm}^{-1}$  to the birefringence  $\Delta\beta = (0.0398 \pm 0.0008) \text{ mm}^{-1}$  [see materials and methods for details (36)]. After the propagation through the waveguide circuit, coincidences between horizontally and vertically polarized photons in the respective right and left waveguides are detected. Figure 3B displays an exemplary HOM dip (8) in the BDC at  $z = 27 \text{ mm}$  that was recorded by varying the arrival time of one photon with respect to the other one: Indistinguishable photons, which arrive with  $\tau = 0$  time delay at the BDC (marked with the red line in Fig. 3B), show suppressed coincidence  $C_{\text{ind}}$  compared to coincidence rates of distinguishable photons  $C_{\text{dis}}$ , which arrive at the BDC with nonzero delays, e.g.,  $|\tau| = 1 \text{ ps}$  (marked with a black line as threshold for distinguishable photons). Figure 3C shows the coincidence rates measured in 12 BDCs with propagation lengths ranging from 6 to 39 mm. The coincidence rates of indistinguishable photons are found to be suppressed independently of the propagation distance, even though the coincidence rates of distinguishable photons oscillate. As a result, we were able to consistently observe a near-perfect HOM visibility  $v = \frac{C_{\text{ind}}}{C_{\text{dis}}} - 1$  of  $-0.969 \pm 0.008$  (Fig. 3D), confirming the predicted propagation-invariant suppression of coincidences. Moreover, the indistinguishable photons that are suppressed in the output  $|0110\rangle$  are found to be bunched in the state  $|0200\rangle$  or  $|0020\rangle$ , as illustrated by the peaks ( $v \approx +1$ ) in the measured HOM traces (Fig. 3, E and F) for these output states.

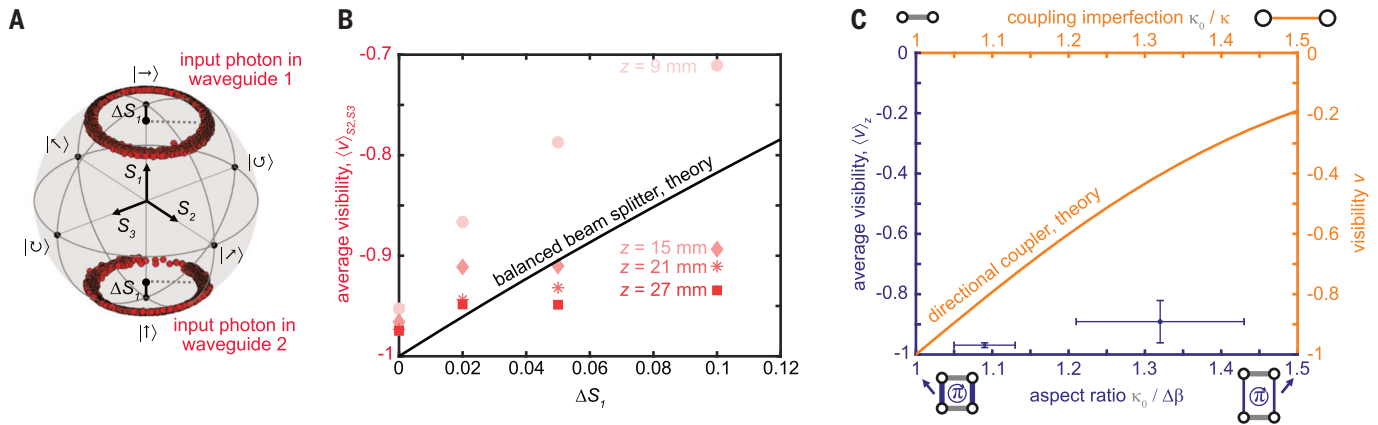
### Topological robustness

We then analyzed the error resilience of the HOM interference in the BDC by comparing the two-photon suppression to the HOM interference in a conventional directional coupler under the influence of similar perturbations. We take into account potential errors in input state preparation as well as structural imperfections affecting the ratio between evanescent coupling and birefringence  $\kappa/\Delta\beta$ . An imperfect input state may be prepared in either the polarization degree of freedom or among the waveguides. Without loss of generality, we restrict the imperfections to the two photons' initial polarization states  $\vec{\psi}_{\text{pol},1} = (1, 1, 0, 0)^T$  and  $\vec{\psi}_{\text{pol},2} = (1, -1, 0, 0)^T$  [represented as normalized Stokes components  $(S_0 = 1, S_1, S_2, S_3)^T$ ] that correspond to the photons' initial horizontal ( $S_1 = 1$ ) and vertical ( $S_1 = -1$ ) polarization, respectively. Thus, perturbations of strength  $\Delta S_1$  yield  $S_1 = \pm(1 - \Delta S_1)$ , and the  $S_2$  and  $S_3$  components acquire random, nonzero values (Fig. 4A) [see materials and methods for details on the experiment (36)]. Therefore, we measure



**Fig. 3. Propagation-invariant suppression.** (A) A horizontally and a vertically polarized photon are launched in the two waveguides of a BDC and subsequently detected in coincidence between the two output waveguides with single-photon counting modules (SPCMs) if the polarization of each photon flipped with respect to the input (the filtering is realized by polarizers). Different realizations of the BDC with lengths of 6 to 39 mm are embedded in a 50-mm-long sample. In their active region, the BDCs comprise two coupled waveguides and, additionally, heavily detuned defect traces serving as stress rods to set the waveguides' slow axes perpendicular to one another. (B) The predicted HOM dip that is the two-photon coincidence rate as a function of photons' delay in arrival time  $\tau$ , is measured in a BDC at a propagation distance  $z = 27 \text{ mm}$ . Indistinguishable photons are available in the center of the dip (shaded red), whereas the photons become indistinguishable outside of the dip, for instance, at a time delay  $\tau = 1 \text{ ps}$  (black line). (C) The coincidence rates of distinguishable (black) and indistinguishable photons (red) as a function of the propagation distance give rise to (D), propagation-invariant suppression indicated by quantum interference visibilities (HOM visibility)  $v = C_{\text{ind}}/C_{\text{dis}} - 1$  ( $v = -1$  for suppression with  $C_{\text{dis(ind)}}$  coincidence rates of (in)distinguishable photons) of  $\bar{v} = -0.969$  and standard deviation  $\sigma = 0.008$  (blue shaded area). (E and F) HOM traces for the output states  $|0200\rangle$  and  $|0020\rangle$ . The coincidences are recorded by splitting the corresponding output after the polarizer with a balanced beam splitter and recording the coincidences among the beam splitters' outputs.





**Fig. 4. Error resilience of two-photon suppression.** (A) A representative measurement of perturbed polarization input state on both photons by  $\Delta S_1$  in the  $S_1 = \pm(1 - \Delta S_1)$  Stokes component shows a randomization in the  $S_2$  and  $S_3$  components. Each red dot represents the measured polarization state within a 20-ms time interval. (B) Launching these perturbed input states in the birefringent directional coupler of propagation distance  $z = 9, 15, 21$ , and  $27$  mm yields average HOM visibilities  $\langle v \rangle_{S_2, S_3}$  [assumed to be measured in

independent combinations for both photons in (A)]. A balanced beam splitter (black) that is subjected to the same perturbations serves as a theoretical benchmark. (C) The measured visibility  $\langle v \rangle_z$  (averaged over all available propagation distances  $z$ , blue) changes slightly as a function of the aspect ratio  $\kappa_0/\Delta\beta$  of the plaquette. Here, a conventional directional coupler with coupling rate  $\kappa$ , which governs suppression when  $\kappa = \kappa_0$ , serves as a theoretical benchmark for the change in HOM visibility  $v$ .

the HOM visibility without knowledge about the actual initial polarization state, which gives rise to averaged visibilities  $\langle v \rangle_{S_2, S_3}$  over all possible  $S_2 - S_3$  combinations for sufficiently long measurement times. Figure 4B compares the measured  $\langle v \rangle_{S_2, S_3}$  for perturbations  $\Delta S_1 \leq 0.1$  and BDCs at four propagation distances to a balanced beam splitter that is subjected to similar perturbations (details provided in the supplementary text). BDCs with very short propagation distances (i.e.,  $z = 9$  mm in our model system) are less capable of providing highly visible quantum interference when dealing with perturbed input states than balanced beam splitters, because the parasitic parts of the input state lead directly to coincidences and imperfect suppression due to the limited correction possibilities in short BDCs. In turn, removing the parasitic parts of the input state (the photons are allowed to initially occupy sites 2 and 3) to the unobserved sites 1 and 4 of the square plaquette by means of single-photon AB dynamics [details provided in the supplementary text (36)] reestablishes perfect suppression for specific propagation distances. Consequently, BDCs with longer propagation distances (i.e.,  $z = 15, 21$ , and  $27$  mm) achieve quantum interference of in general higher visibility than the balanced beam splitter. The optimal performance of the BDC occurs at  $z = 27$  mm (at the maximum of suppressed photon pairs in Fig. 3C), for which the average visibility becomes almost independent on the perturbation strength.

To test the latter issue, a directional coupler with  $L = \frac{\pi}{4\kappa_0}$  serves as a reference. With precisely matched coupling rate ( $\kappa = \kappa_0$ ), such a component exhibits perfect suppression, where as any deviations from this design point cause a

rapid deterioration of the visibility of the HOM dip, with  $v$  exceeding the threshold of  $v = -0.5$  for relative deviations of  $\kappa_0/\kappa$  on the order of 1.3. By contrast, even when we intentionally modify a BDC by adjusting the birefringence to  $\Delta\beta = (0.0329 \pm 0.012) \text{ mm}^{-1}$ , resulting in a heavily unbalanced configuration with  $\frac{\kappa}{\Delta\beta} = (1.32 \pm 0.11)$ , the mean visibility over the different propagation distances is only marginally affected [ $v_z = (-0.89 \pm 0.07)$ , Fig. 4C], again highlighting the error resilience of topological q3 quantum interference. Likewise, the fact that  $v$  remains independent of the propagation distance  $z$  is grounded in the topological character of the system of the two-photon interference [details are provided in the supplementary text (36)].

### Concluding remarks

Our findings pave the way for harnessing topological features in the next generation of quantum photonic architectures with superior stability and resilience against imperfections. We anticipate that photonic devices using photonic degrees of freedom (43) to introduce topological protection in many-particle dynamics will find use cases in a broad range of new quantum technological applications alongside non-Abelian geometric phases (44), non-conservative systems (45, 46), and many-photon quantum states (47) with interactions (48, 49).

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cowrote the manuscript. **Competing interests:** The authors declare that they have no competing interests. **Data and materials availability:** All experimental data and any related experimental background information not mentioned in the text can be found at the Rostock University Publication Server repository (50). **License information:** Copyright © 2024 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.sciencemag.org/about/science-licenses-journal-article-reuse>

SUPPLEMENTARY MATERIALS

[science.org/doi/10.1126/science.ado8192](https://science.org/doi/10.1126/science.ado8192)  
Materials and Methods  
Supplementary Text  
Figs. S1 to S4  
References (51, 52)

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